

TOPIC 8. CERAMIC MATERIALS I II: PROCESSING AND APPLICATIONS OF CERAMIC MATERIALS

- **Processing of ceramics :**
 - Powder manufacturing
 - Powder conditioning
 - Shaping-forming
 - Drying and sintering
- **Processing of glasses**
- **Applications and Examples**

Ceramic Materials



Pottery (clay)



Duralex (glass ceramic)



Pyrex (Borosilicates)



Bricks

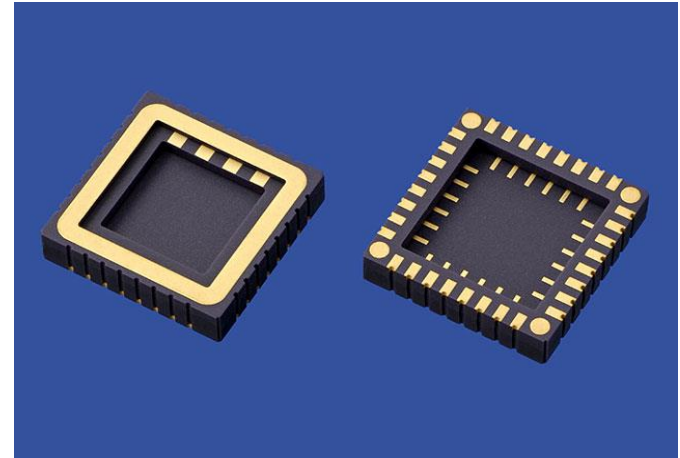


Refractory ceramics

Ceramic Materials



Nick Fury's car window (tempered glass)



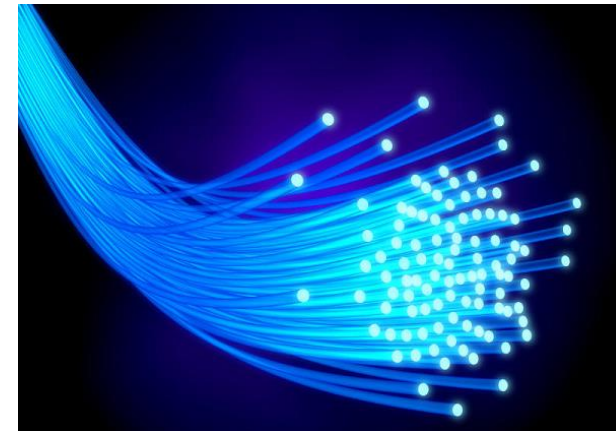
Microelectromechanical systems (MEMS)



Thermal insulator (glass wool)

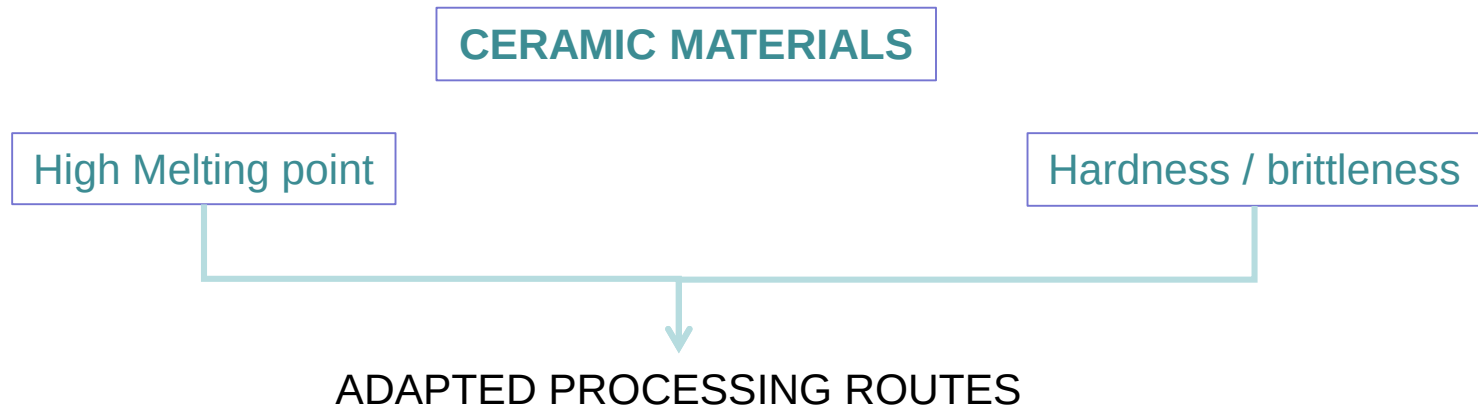


Ceramic braces



Optical fiber (SiO₂)

Introduction



Ceramic classification, regarding final properties:

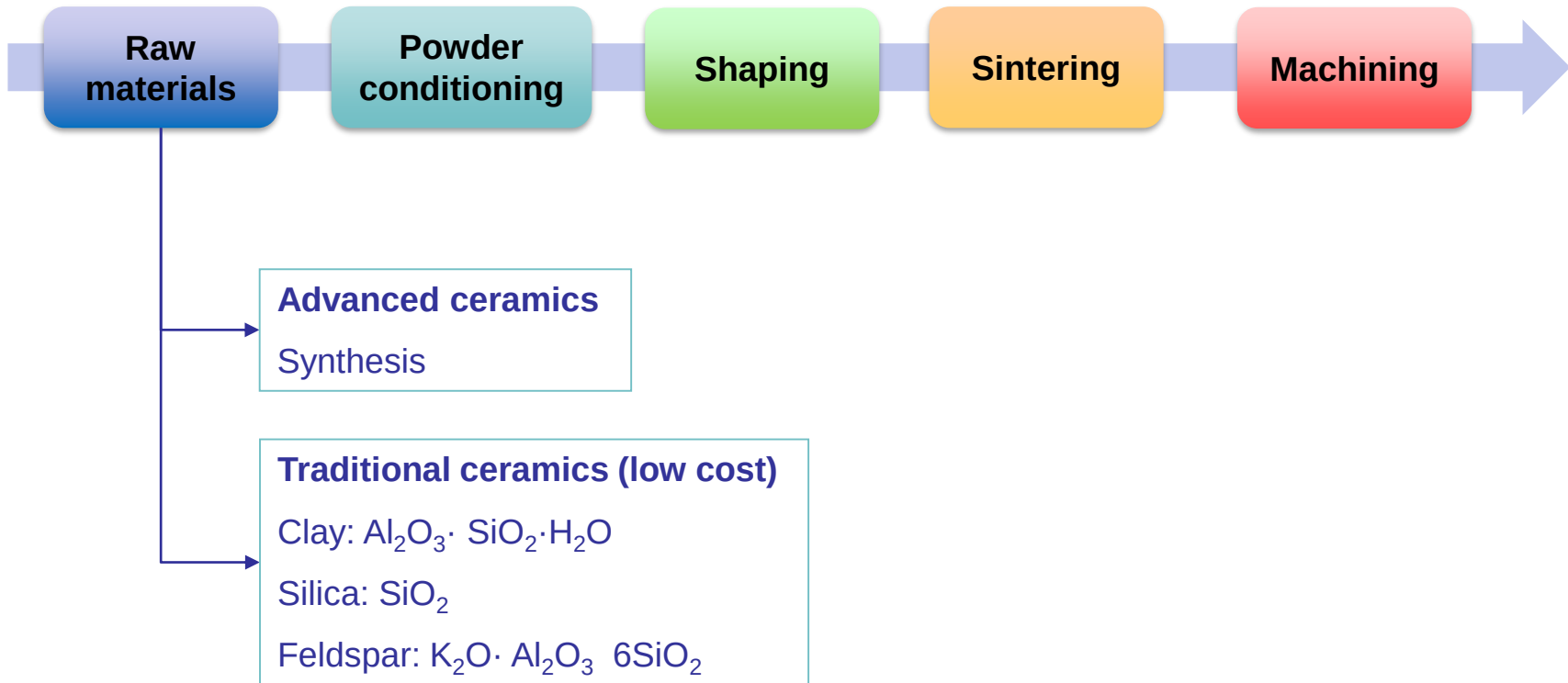
Traditional ceramics

Advanced / technical ceramics

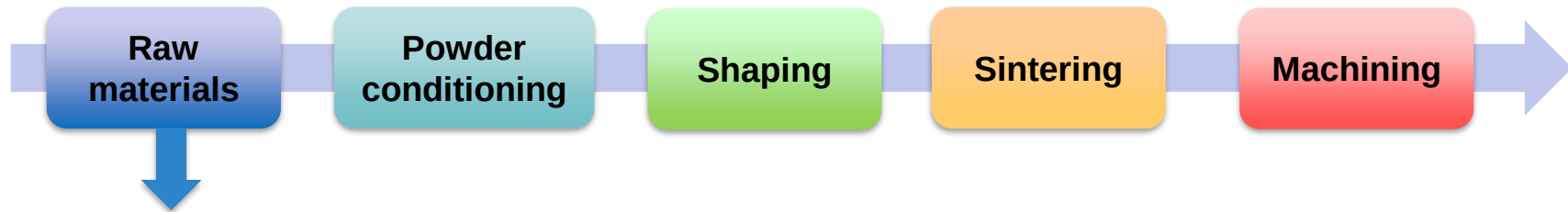
- oxides
- non-oxides

Glasses

PROCESSING OF CERAMICS



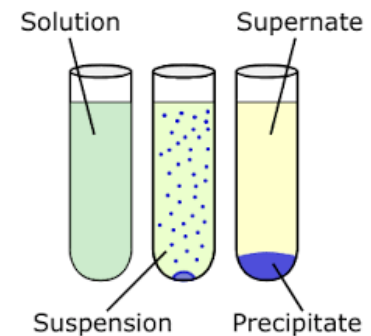
Processing of ceramics. Powder manufacturing



Advanced (structural) ceramics powder synthesis

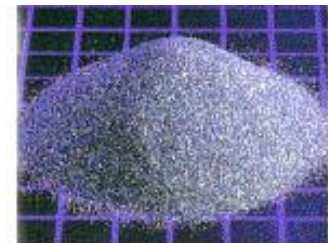
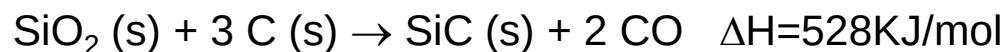
- Oxidation/reduction reactions
- Thermal decomposition
- Liquid-solid reactions
- Solid-solid reactions
- Hydrothermal synthesis
- Liquid Phase Synthesis: precipitation, hydrolysis, evaporation
- Other methods

Obtain ceramic powder



Example: Synthesis of SiC

- **Acheson Process:** T: 1600-2500°C (electric resistance furnace, 36h)



PROCESSING OF CERAMICS: Powder conditioning

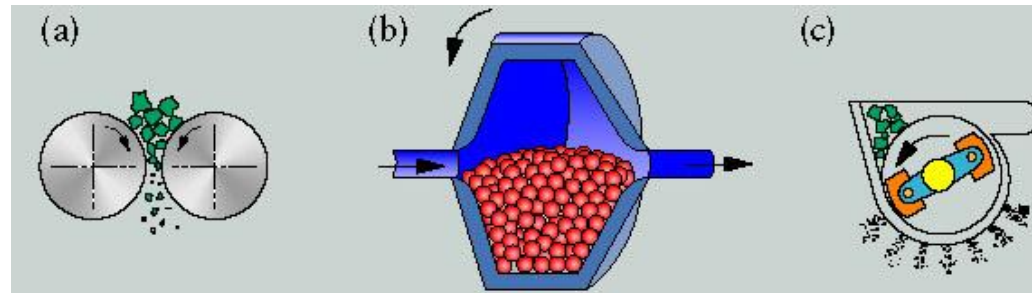


Mixing with additives

Agglutinants
Lubricants
Wetting agents
Plasticizers
Deflocculents
Granulate

Milling (**wet or dry**) to get small particles:

- (a) Roll crushing (dry)
- (b) Ball milling (wet or dry)
- (c) (Hammer milling (dry).



PROCESSING OF CERAMICS: Powder conditioning

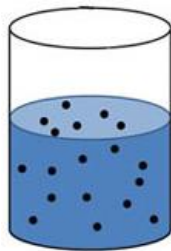


Mixing with additives

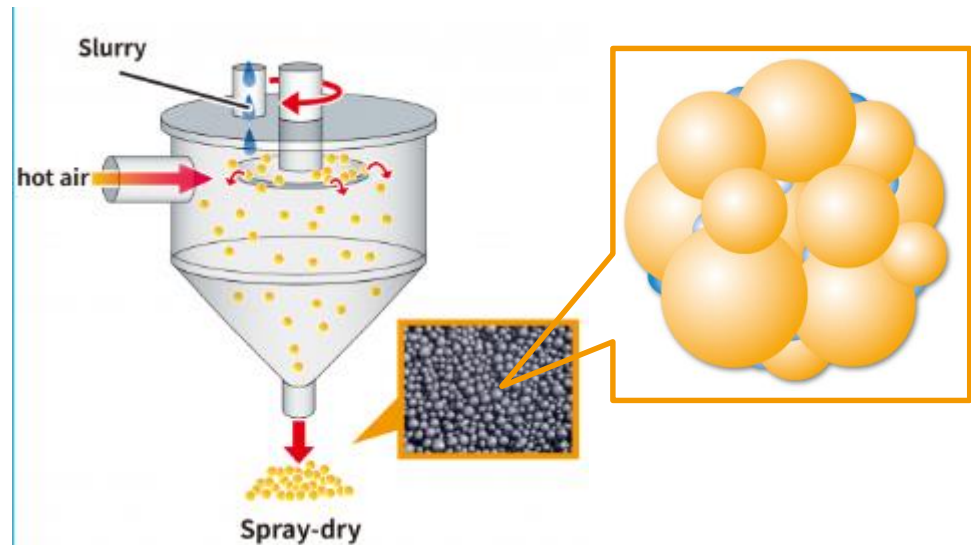
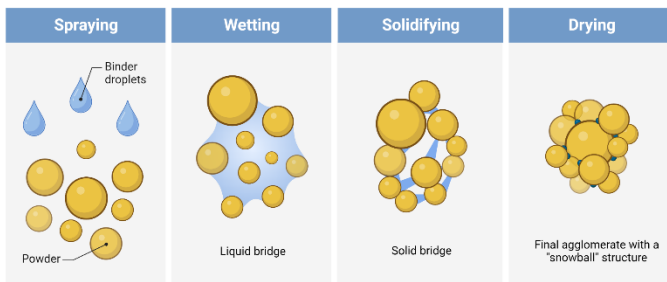
Granulation

Suspensions/Slurries (60-70% solid, agglomerates and water) with adequate viscosity are:

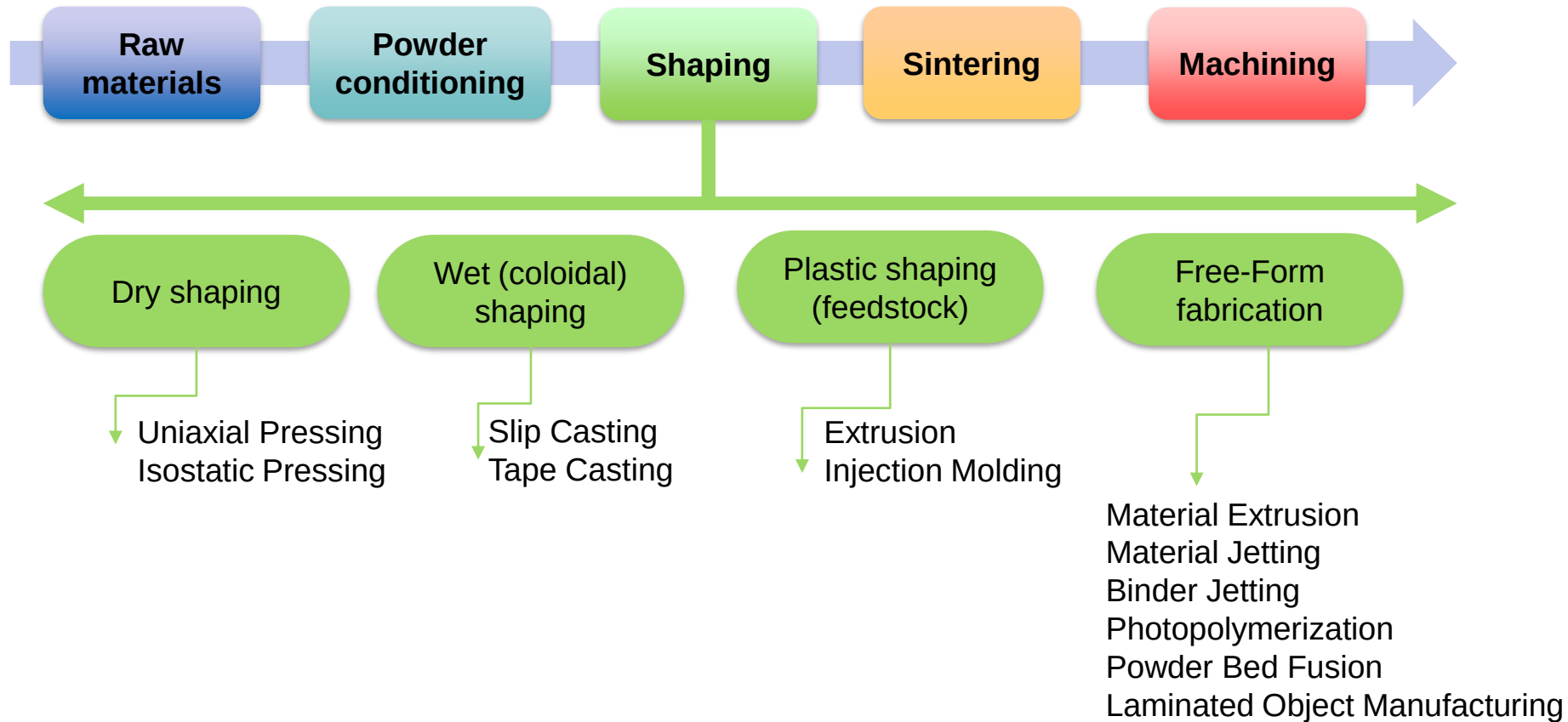
- Pulverized
 - Dried with hot gases
- } ⇒ **GRANULATED POWDER**
(spherical particles, pellets)



Suspension



Processing of ceramics. Shaping

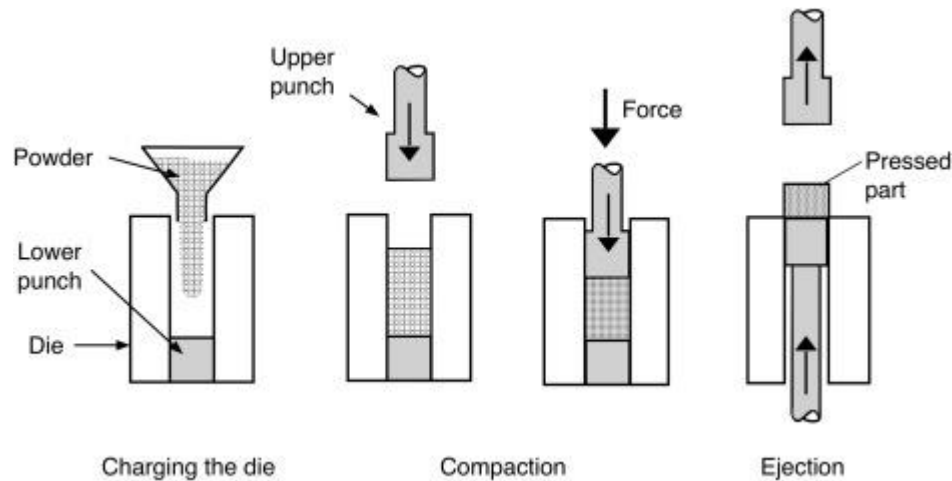


Processing of ceramics. Shaping

Shaping

Dry shaping

Uniaxial Pressing (Pressing)



- Good for complicated pieces and ↑ production velocity
- Variables to control:
 - good filling
 - good compressibility
 - use coagulants/additives/agglomerates



Ceramic nozzles (complete and half)



Alumina (Electrical components)

UNIAXIAL PRESSING

https://www.youtube.com/watch?v=4zXkn9iyC0c&ab_channel=GKNPowderMetallurgy



Processing of ceramics. Shaping

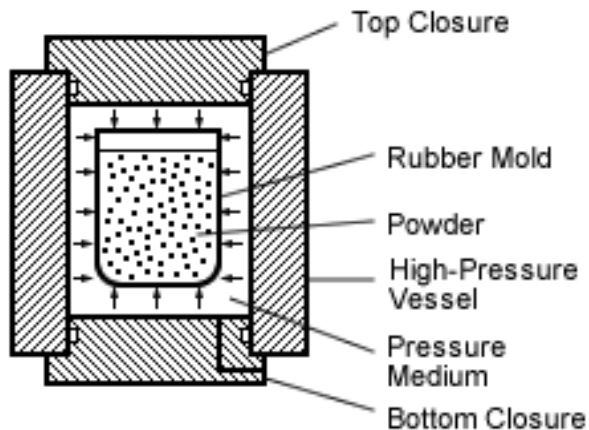
Shaping

Dry shaping

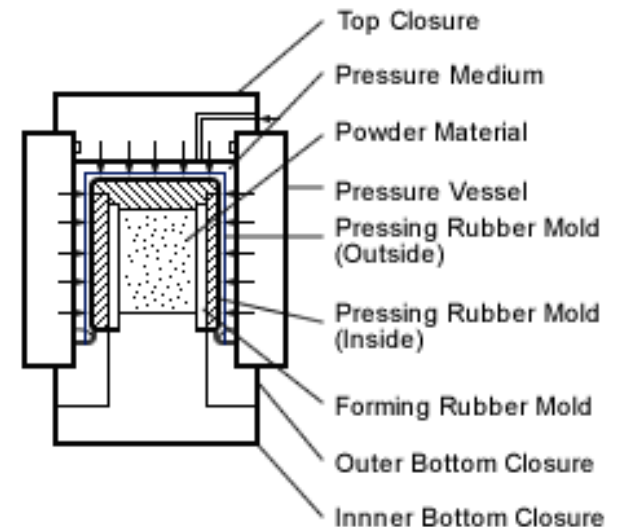
Cold Isostatic Pressing (CIP)

Application of **hydrostatic pressure** using a fluid on dry ceramic powder

➤ **Wet Bag** Pre-shape sample, load in “flexible mold” (silicone, polyurethane...), (vacuum), immerse in hydraulic fluid, apply pressure.



➤ **Dry Bag** Load powder in thick rubber bag, with rigid support, then apply pressure (easier for automation)



1. Processing of ceramics. Forming-Shaping

Shaping

Dry shaping

Cold Isostatic Pressing (CIP)

Advantages:

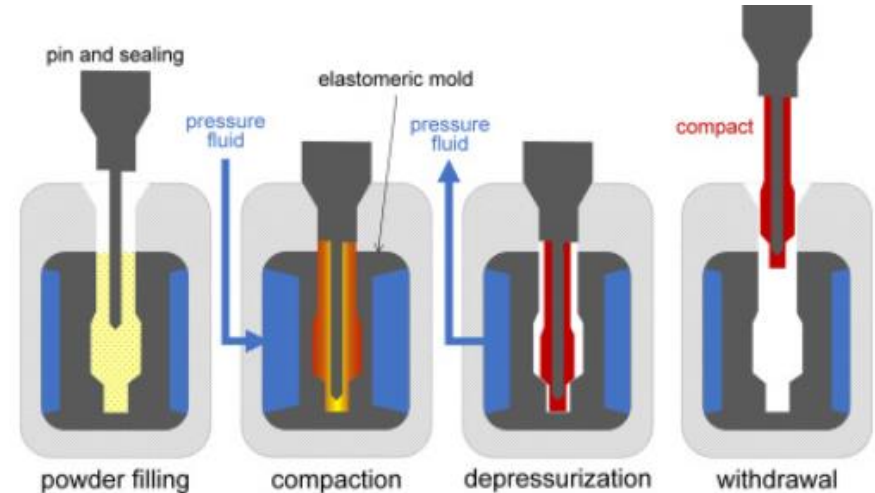
- no required granulated powders
- Uniform components
- No ejection cracks
- No lubricant addition
- High green strength (possible machining)
- Low porosity
- Uniform density and properties

Drawbacks:

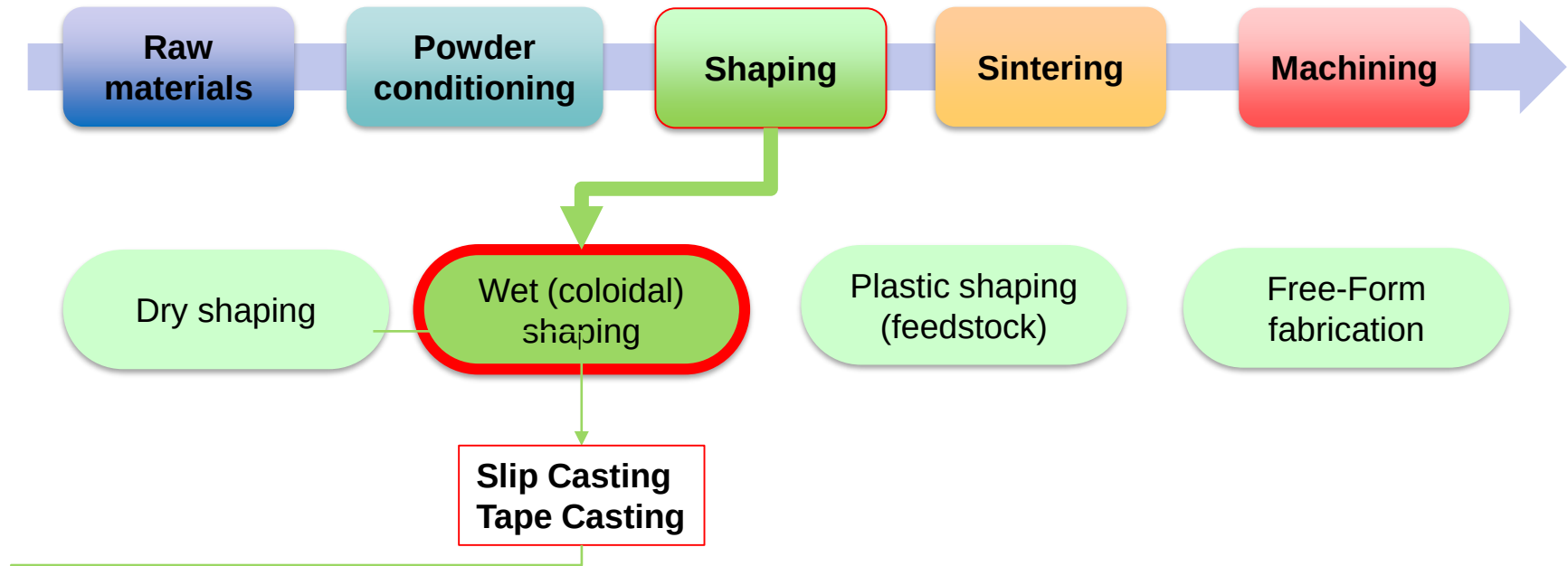
- Slow and discontinuous process
- Low dimensional accuracy
- Low surface finishing

Application (dry bag):

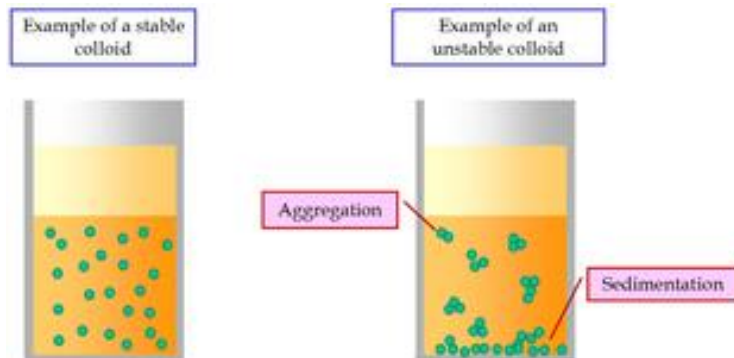
- Insulating part of spark plugs



Processing of ceramics. Shaping



- Stable **colloidal suspensions (slips)** are used for forming/ shaping of ceramic parts
- Slip: suspension of a *colloidal* ceramic particles in an *immiscible liquid*, generally water



Slip characteristics:

- Chemically inert
- Optimum flowability, low viscosity
- Maximum density (as possible)
- Detachment ability (from the mould).
- Low shrinkage during drying.
- Low turbulence flow (newtonian)

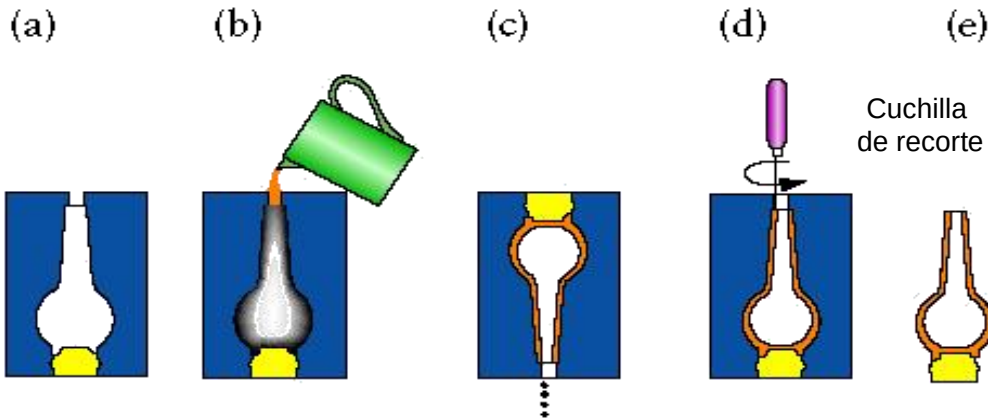
Processing of ceramics. Forming-Shaping

Shaping

Wet shaping

Slip Casting

Plaster or alabaster porous moulds



- Porous mold (plaster)
- Pouring of slip into porous mold. Mold absorbs some of the water
- Remaining suspension is poured out
- Trimming
- Part is removed and ready for firing

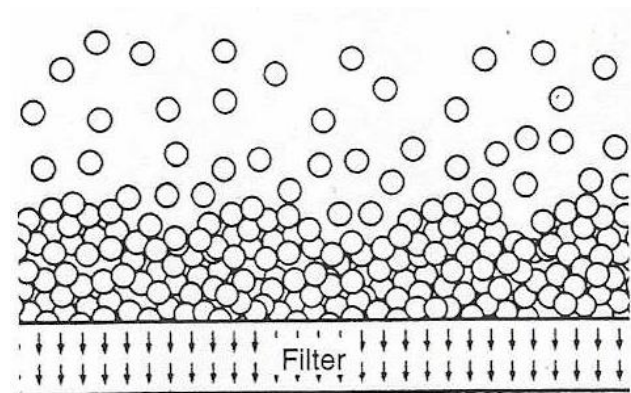
Bathroom, plant pots, fountains

Advantages

- Big-size components
- Complex shapes
- Cold processing
- Low-cost equipment

Drawbacks

- Low production rate
- Maximum thickness: 10 mm
- Low dimensional accuracy



Deposition of ceramic powder into porous mold and capillary extraction of carrier fluid

1. Processing of ceramics. Forming-Shaping

Shaping

Wet shaping

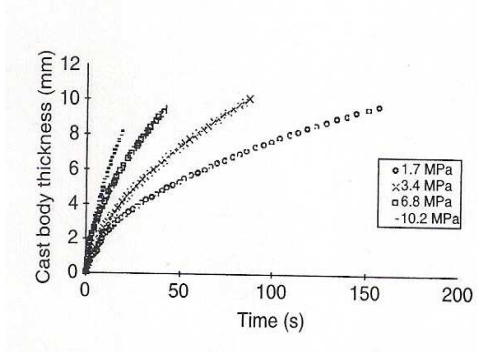
Slip Casting



VARIANT: Slip casting with pressure

↓↓↓ processing time

↑↑ thickness of components reliability



Wall thickness as a function of casting time for pressure casting at different applied pressures (SiC)



SLIP CASTING

https://www.youtube.com/watch?v=FZzOTX9lhqs&ab_channel=ShaneeLouw



1. Processing of ceramics. Forming-Shaping

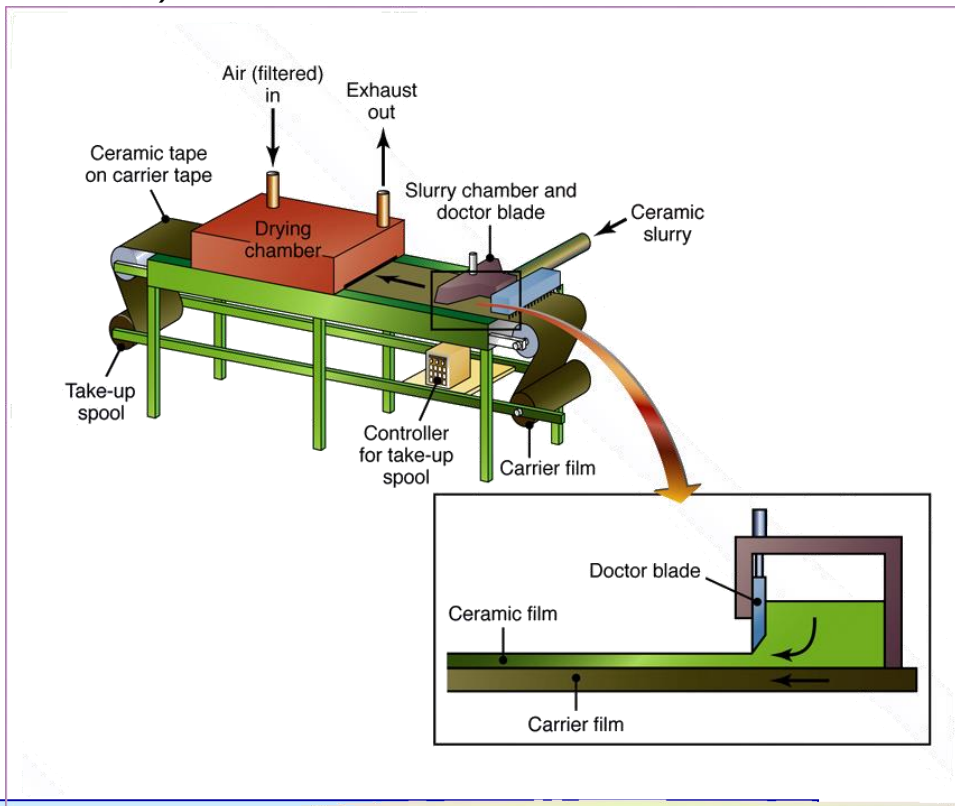
Shaping

Wet shaping

Tape Casting (Doctor Blade)

A ceramic slip (colloidal suspension) is cast onto a flat moving belt.

Fabrication of ceramic layers/ Ceramic sheets (thickness lower than 1.5 mm)

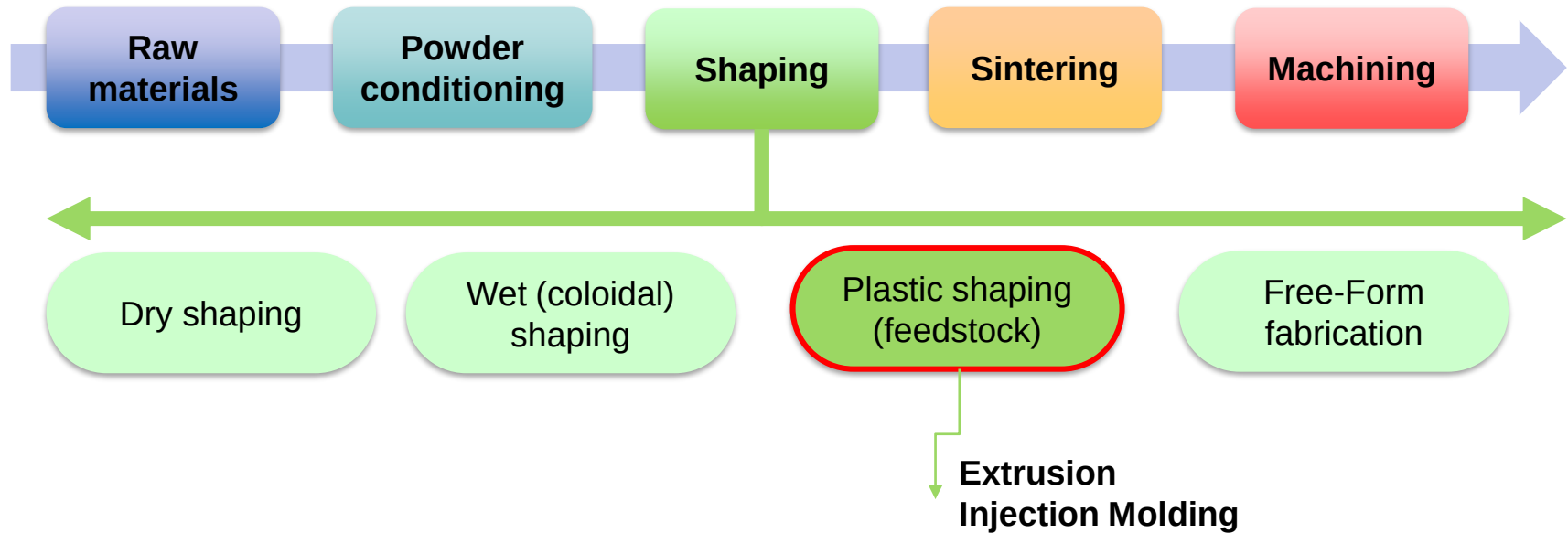


- Continuous process
- Acceptable accuracy

Applications:

- **Multilayer capacitors** (with a metallic conductor phase, Ag, Au, Cu, ...). One-process sintering
- Fuel cell components**

Processing of ceramics. Shaping



Ceramic powder is mix with a binder (wax, thermoplastic, additives etc.) in order to produce a **feedstock**.



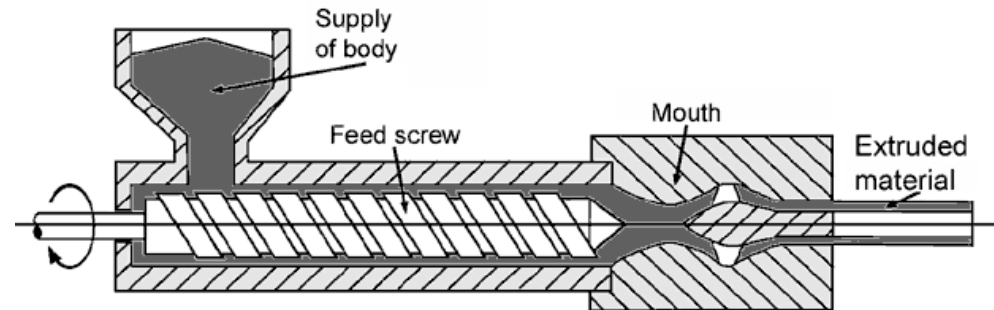
1. Processing of ceramics. Forming-Shaping

Shaping

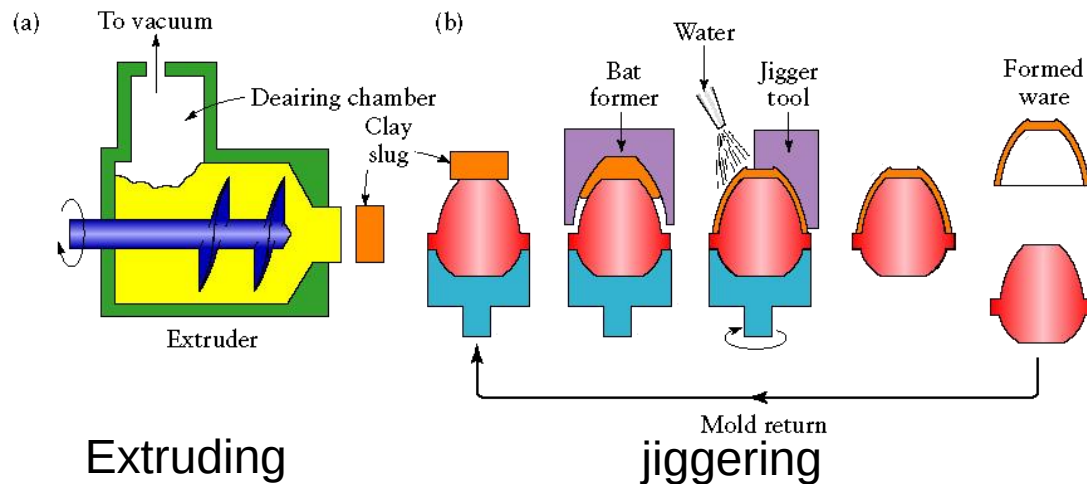
Plastic shaping

Extrusion

- Feedstock is feed into extruder hopper.
- Feedstock is forced through the shaping die by a ram, screw, or piston
- Extruded material emerges as product with **uniform cross-section**



VARIANT: EXTRUSION AND JIGGERING



- Extrusion of ceramic component slug
- Moulded as bat on a revolving mould



1. Processing of ceramics. Forming-Shaping

Shaping

Plastic shaping

Extrusion (characteristics)

- Small section components (but above 15 mm in smallest dimension)
- High reproducibility for constant sections
- Axisymmetric shapes
- Appropriated for technical ceramics: alumina, zirconia, silicon nitride, silicon carbide, ...

Applications:

- **Refractory tubes:** for thermocouples, furnaces, catalysts, bricks (traditional ceramics...



Advantages

- Hollow, small-diameter components.
- High production rate.
- Cold processing
- Low-cost equipment

Drawbacks

- Constant transverse section
- Limited thickness



Processing of ceramics. Forming-Shaping

Shaping

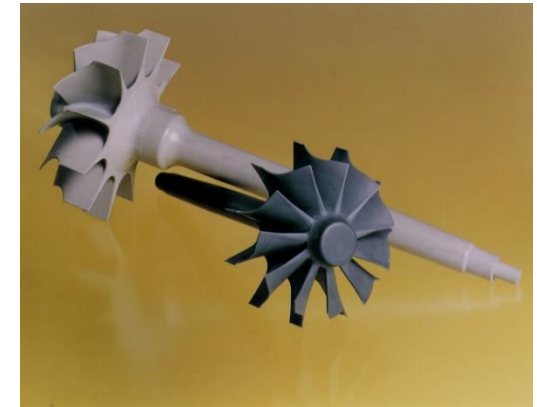
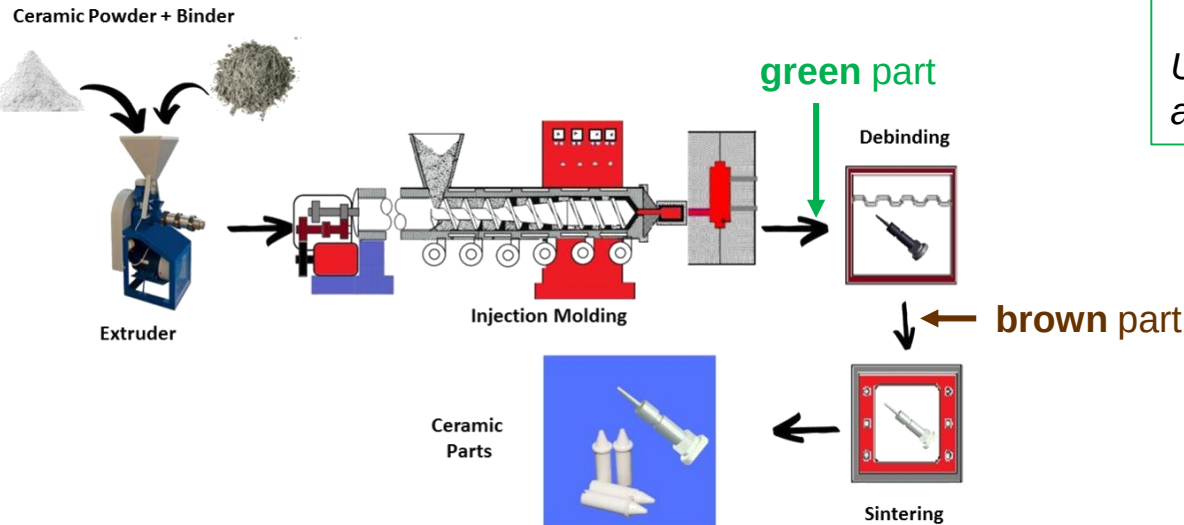
Plastic shaping

Injection Molding

1. Ceramic powder is mixed with binder or wax to get **feedstock**
2. Feedstock is forced through the shaping die by a ram, screw, or piston
3. Injection molded material emerges as product (**green** part)
4. Binder is removed by pyrolysis/thermal evaporation/solvent washing (**brown** part)
5. Part is *sintered*

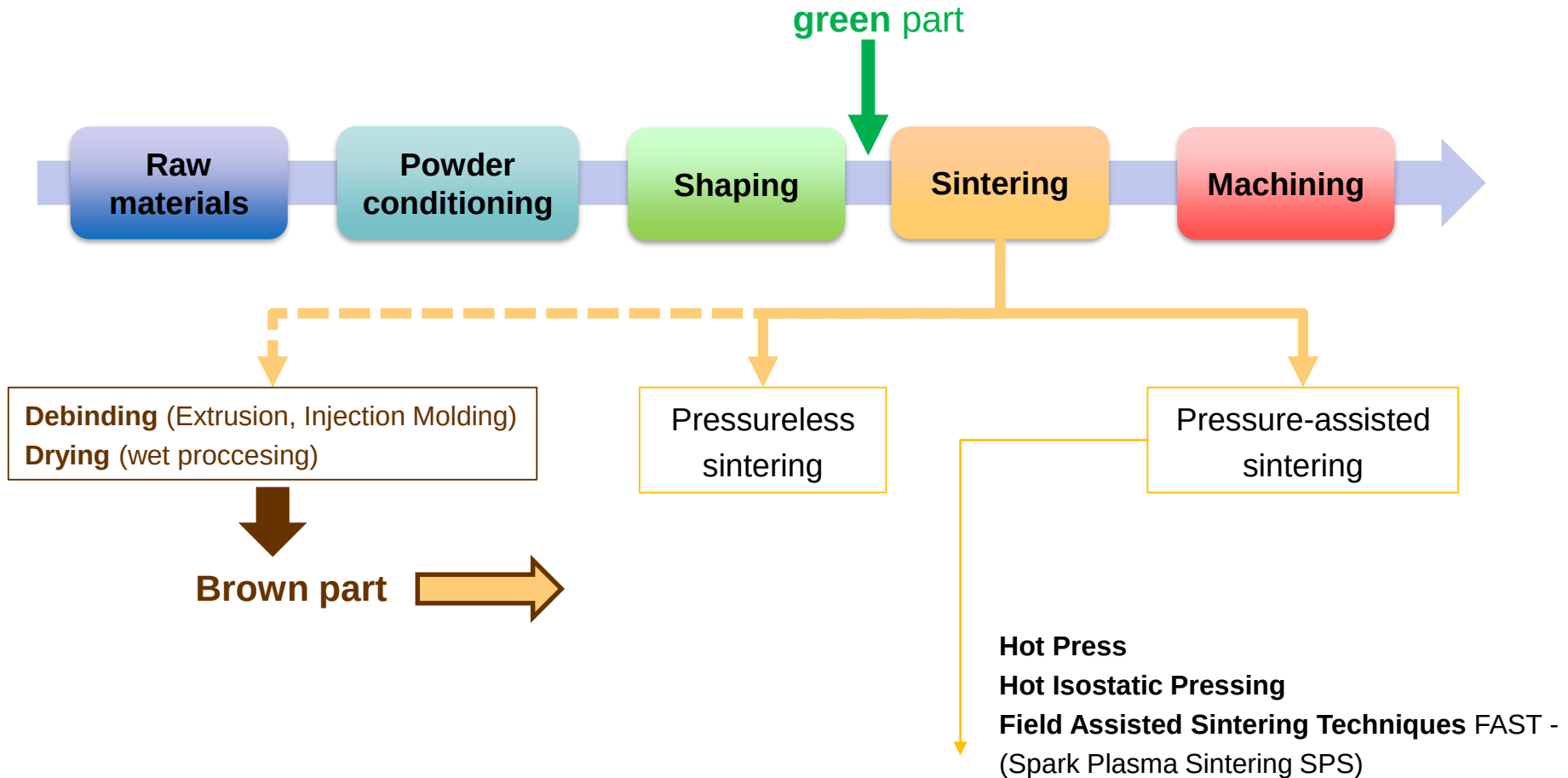
- Good accuracy
- Expensive

Used for high-tech application such as rocket engine components

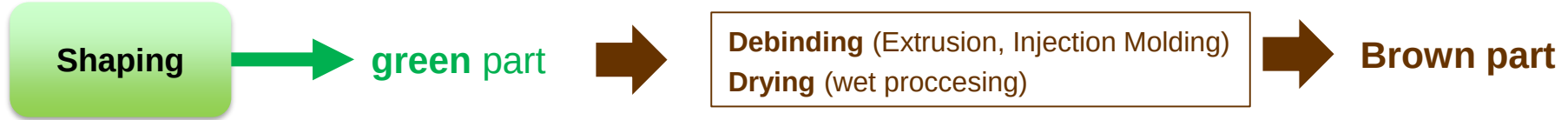


Injected-molded turbo charger (in green and sintered state)

PROCESSING OF CERAMICS



Processing of ceramics. Drying and Sintering



Important stage: Cracking, distortion or warping can occur $\Rightarrow$ slow heating and atmospheric control is necessary

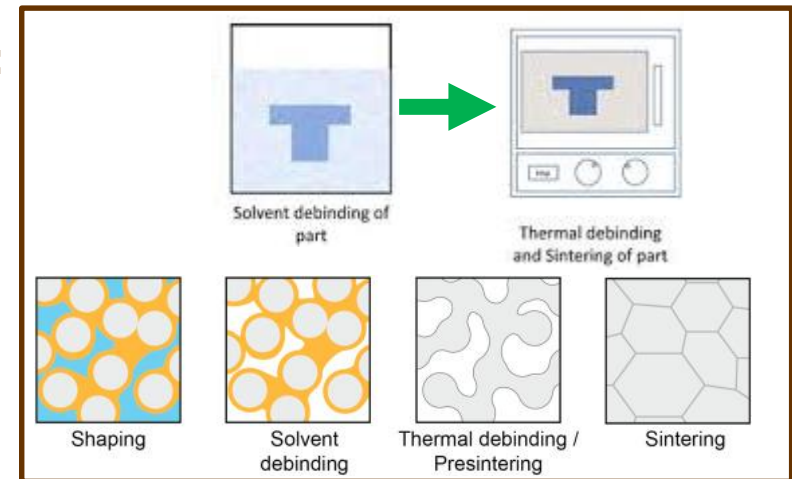
Shrinkage: 15-20 % shrinkage can occur during drying/debinding

Debinding (Extrusion, Injection Molding):

Removal of polymeric binder

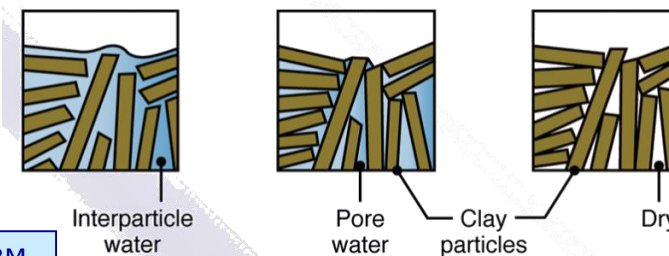
Usually in two steps:

- Solvent debinding
- Thermal debinding



Drying (Wet processing):

Removal of water



Processing of ceramics. Sintering

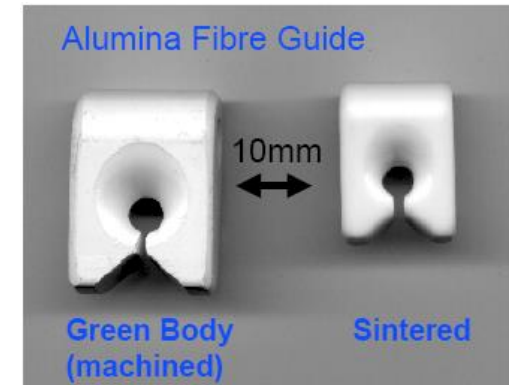
Sintering

Heating the compact part to a high temperature **below the melting point**, where strong bonds develop between particles though **diffusion processes** $\Rightarrow$ ceramic part gets its strength and hardness.

Densification and **shrinkage** occurs

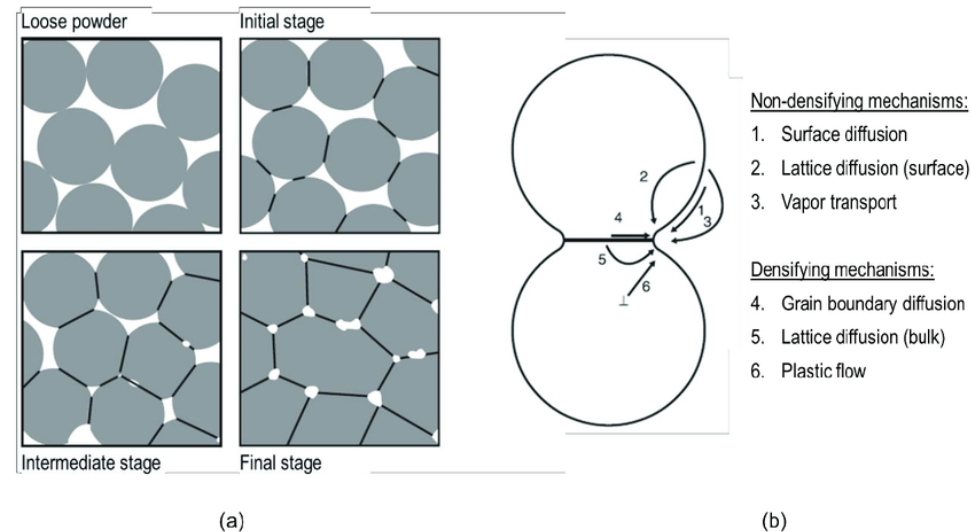
Driving force for sintering $\Rightarrow$ lower surface energy

- Curvature of particle surface
- External applied pressure
- Chemical Reaction



Material Transport Mechanism

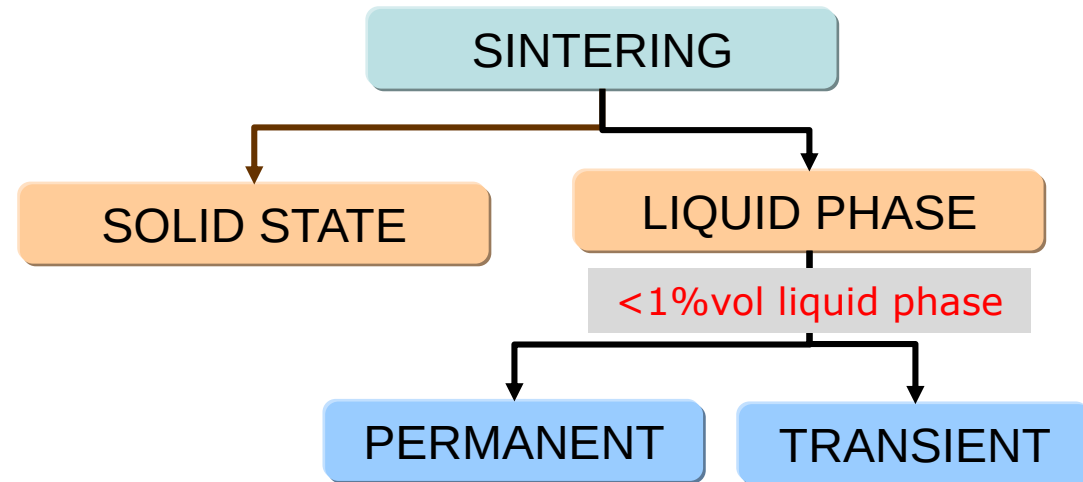
Transport Mechanism	Material source	Material sink
Surface diffusion	Grain surface	neck
Lattice diffusion	Grain surface	neck
Vapor Transport	Grain surface	neck
Grain boundary diffusion	Grain boundary	neck
Lattice diffusion	Grain boundary	neck
Plastic flow	Bulk grain	neck



Processing of ceramics. Sintering

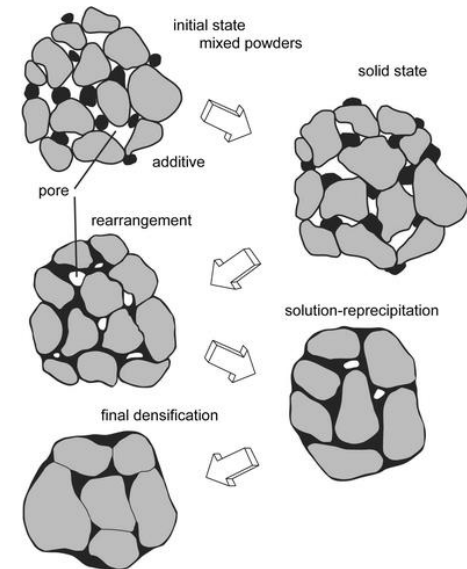
Factor affecting sintering

- Temperature
- Time
- Green density
- Uniformity of green density
- Atmosphere
- Impurities
- Size distribution
- Particle size



Liquid phase sintering: a small amount of liquid phase enhances the diffusional mechanisms that drive densification.

- Rapid initial particle **rearrangement** due to capillary forces,
- Mass transport through liquid is generally orders of magnitude faster than through solid,



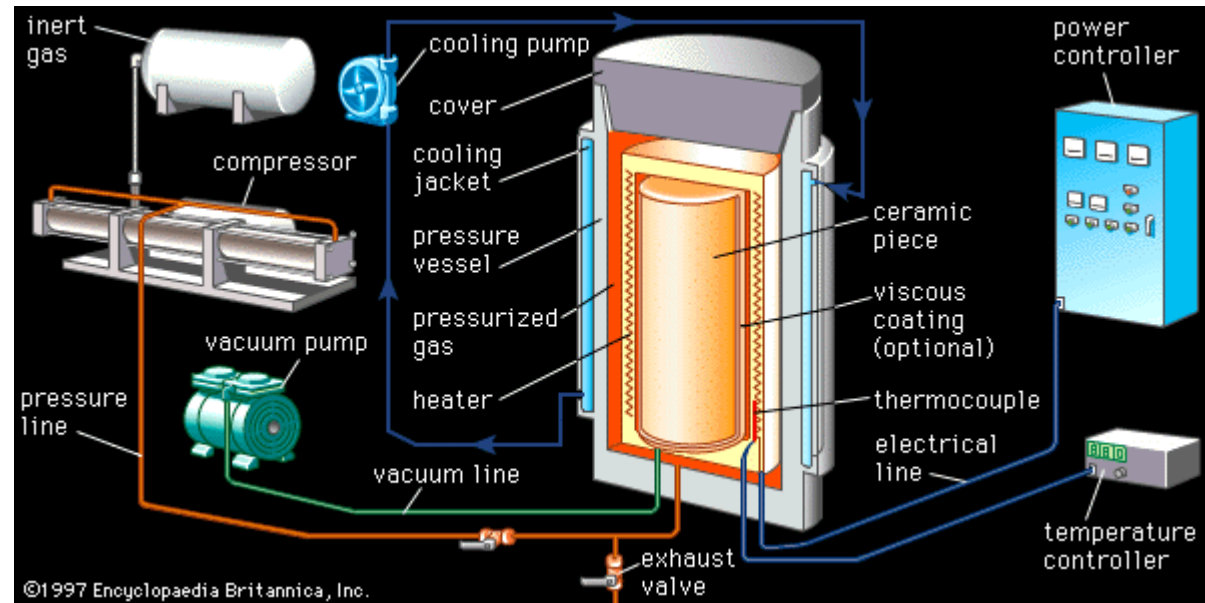
Processing of ceramics- Pressure Assisted Sintering

Hot Isostatic Pressing (HIP)

Fluid = inert gas

Stages:

1. Encapsulating ceramic powder in metallic mould
2. Vacuum of metallic mould
3. Mould in furnace (autoclave) of $\uparrow P$ (up to 15 bars) and T (up to 2000°C)



Advantages:

- Very good pressure distribution $\Leftrightarrow$ uniform properties.
- Versatility of shapes
- For high hardness (mechanical properties) components

Drawbacks:

- Very-high cost of furnace
- Scarce dimensional accuracy
- Slow, discontinuous process.
- Inert gas consumption

Processing of ceramics- Pressure Assisted Sintering

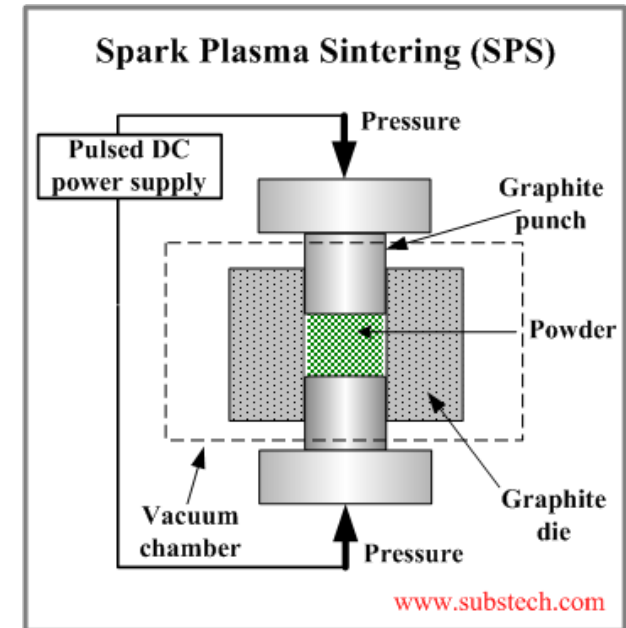
Field Assisted Sintering Techniques FAST - (Spark Plasma Sintering SPS)

Powders are placed in a mold (usually graphite), and pressure is applied. An **electric current** is **pulsed** directly through the mold and/or sample while a pressure is applied. Joule heating heats the sample directly.

- High heating rates (up to 1000 K/min)
- Low processing cycle times(minutes).
- Lower temperatures and mould pressures compared to conventional techniques

Advantages:

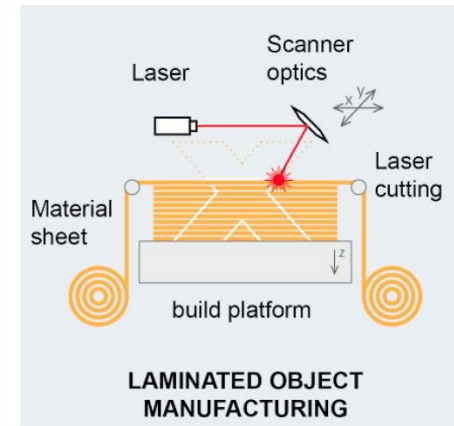
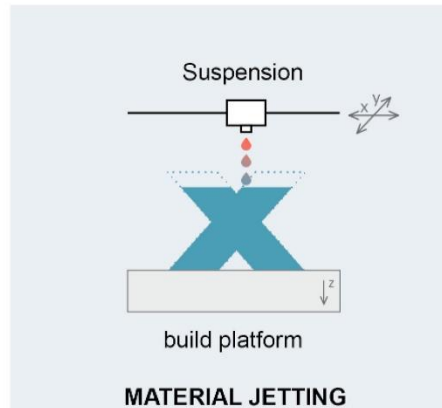
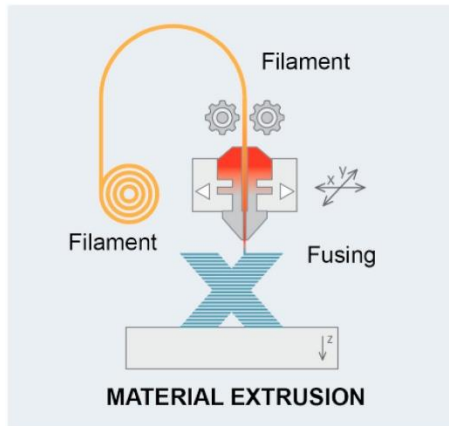
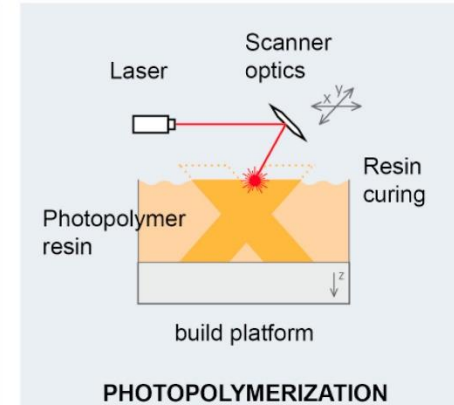
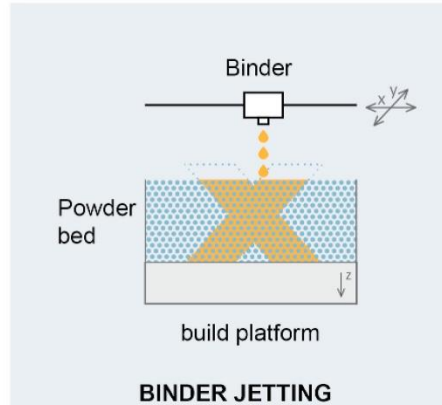
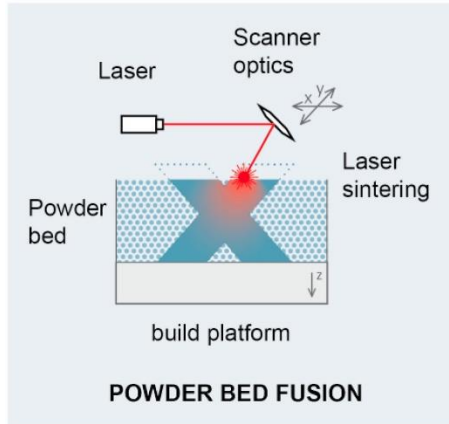
- Fast sintering process.
- Uniform sintering
- Different materials can be processed
- Low grain growth



Drawbacks:

- Simple symmetrical shapes
- Slow, discontinuous process.
- Expensive

Processing of ceramics- Additive Manufacturing of Ceramics

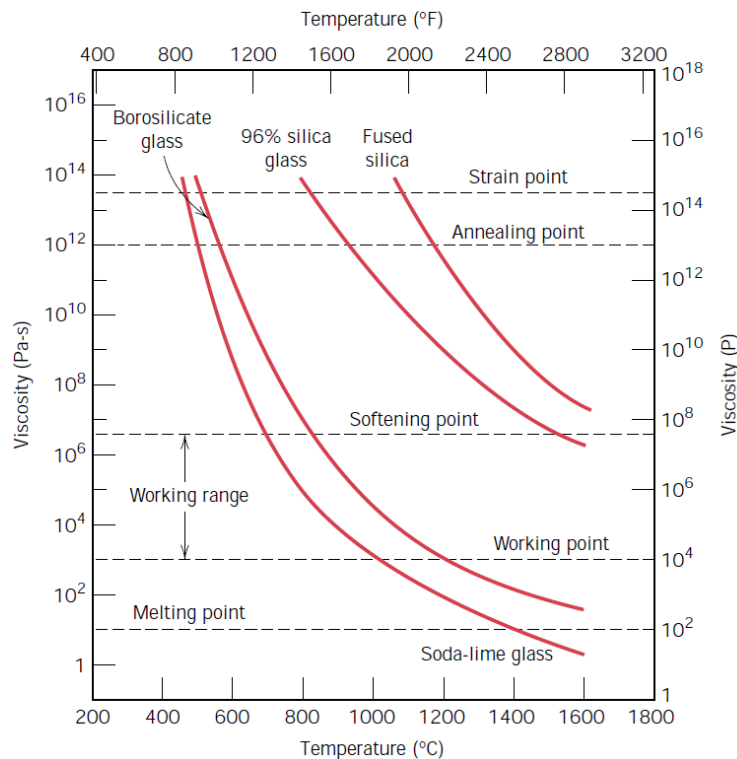


Glasses. Characteristics

- Amorphous materials with different compositions (eg, SiO_2 or B_2O_3).

Undercooled liquid: viscosity increases when temperature decreases

Viscosity, $\eta = f(\text{Temperature, composition, thermal history})$



- **Strain point:** $\eta = 3 \cdot 10^{14} \text{ P}$
- **Annealing point:** $\eta = 10^{13} \text{ P}$
- **Softening point:** $\eta = 4 \cdot 10^7 \text{ P}$
- **Working point:** $\eta = 10^4 \text{ P}$
- **Melting point:** $\eta = 100 \text{ P}$

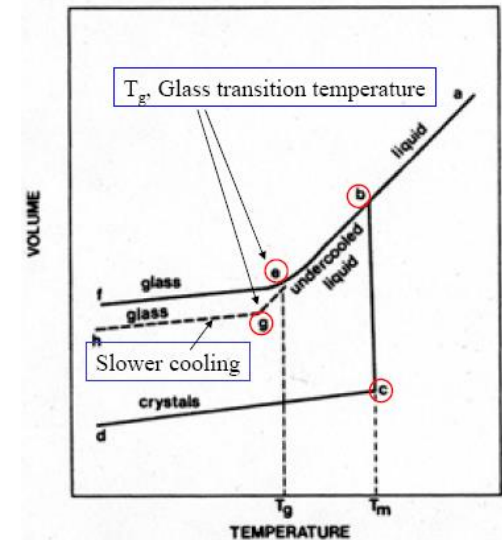


Fig. 18.2 Volume-temperature characteristics (schematic) for crystals, liquid and glass

T_g : glass transition temperature

$T_g = f(\text{cooling rate})$

Processing of Glass

Glass Processing: Increasing temperature above T_g and shaping

Glass products:

1. Flat sheets or plates

window glass, glass doors etc

2. Rods or tubing

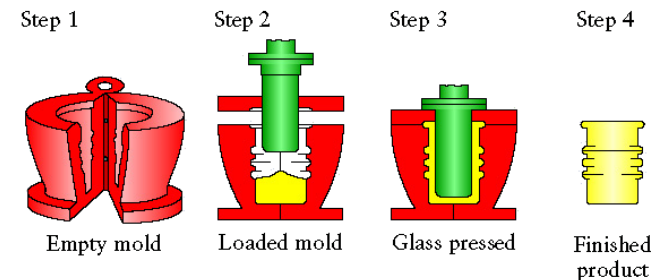
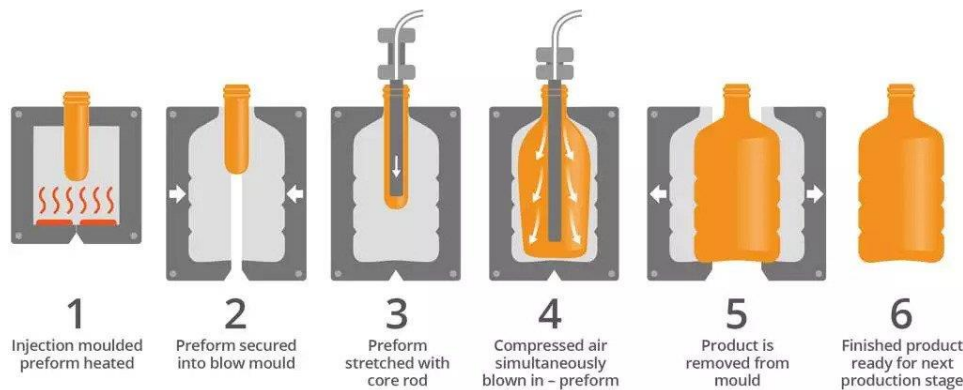
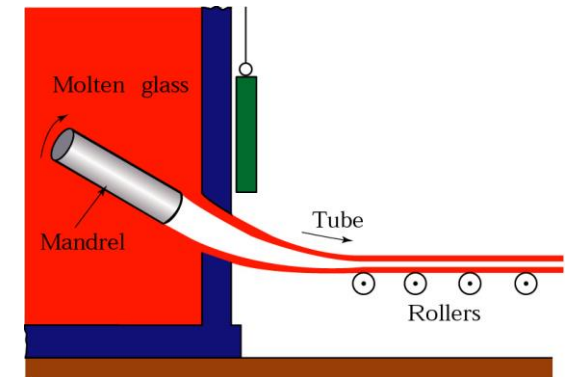
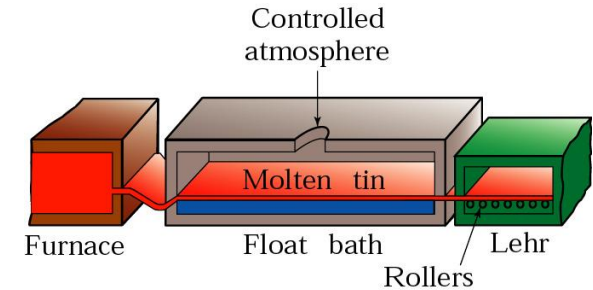
Neon lights, chemicals decorative

3. Discrete products

Bottles, vases, headlights etc

4. Glass fibers

To reinforce composite materials, or for thermal insulating material



GLASS BOTTLE PRODUCTION

https://www.youtube.com/watch?v=A_M8WBJMcM0&ab_channel=DiscoveryUK



Glass processing. Tempering thermal treatment

TEMPERED GLASS

- Glass heated **above** T_g but below the softening point and **cooled rapidly**
- Surface cools to below T_g first and contracts.
- Interior cools next and contracts drawing the exterior into compression.
- Tensile stresses develop in the interior and compressive stress on the surface.
- **Compression stresses in the surface** to increase strength. Avoid growth of surface cracks.
- Examples: Auto side windows and safety glasses.

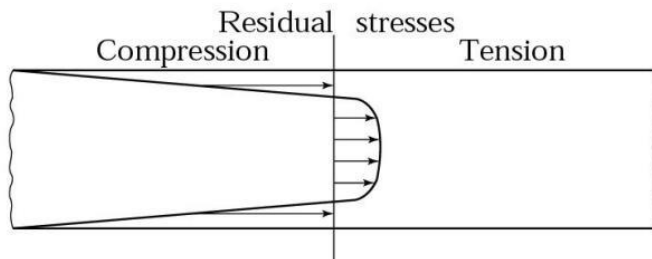
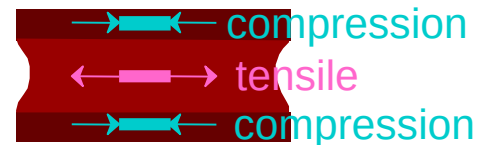
Hot glass



Cooled surface



Final cooling (RT)



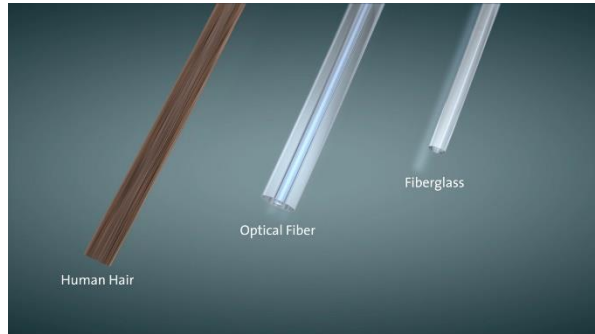
TEMPERED GLASS

<https://www.youtube.com/shorts/YNIzDtvHlu4>

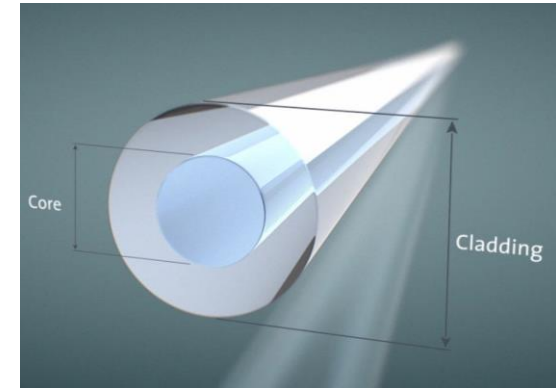


Glass processing. Glass Fibers

GLASS FIBERS



optical glass fiber



Core of a glass optical fiber: Silica (SiO_2) doped with Germanium dioxide (GeO_2) and alumina (Al_2O_3) to *increase* refractive index
Cladding: Pure Silica or Silica doped with fluorine or boron to *decrease* refractive index



Glass wool: used for thermal insulation

Glass wool (short glass fibers 20-30 mm): made by a *centrifugal casting* process where molten glass is ejected (spun) from a rotating head through multiple orifices



glass fiber

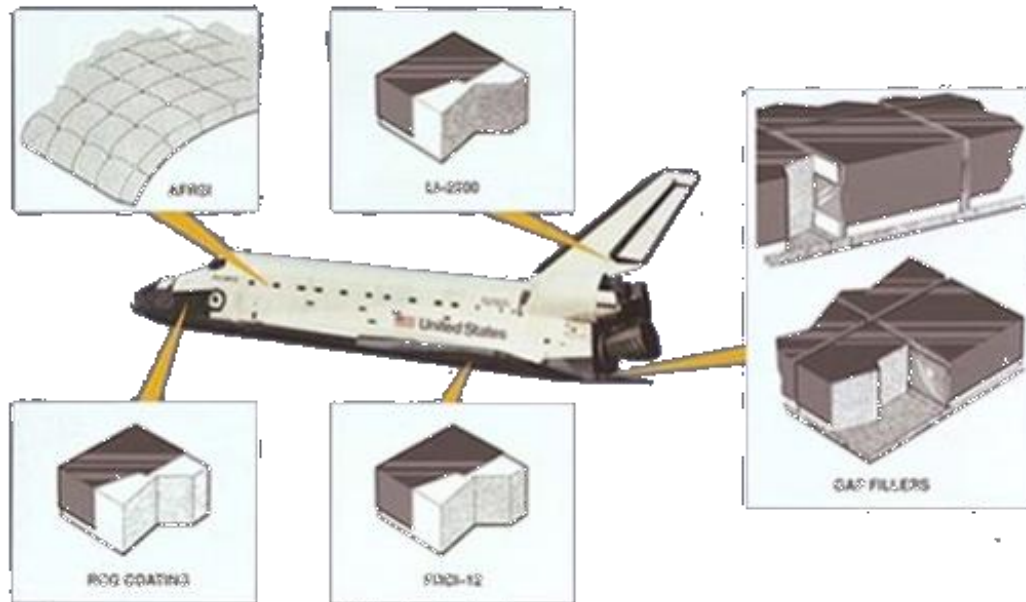
APPLICATIONS OF ENGINEERING CERAMICS

Thermal protection System of space shuttle

Specific material for each position in the shuttle

Material used

- Ceramic tiles (HRSI) (LRSI) (*FRCI*)
- C reinforced with C RCC
- Felt covered with silicone (*FRSI*) (*FIB/AFRSI*)

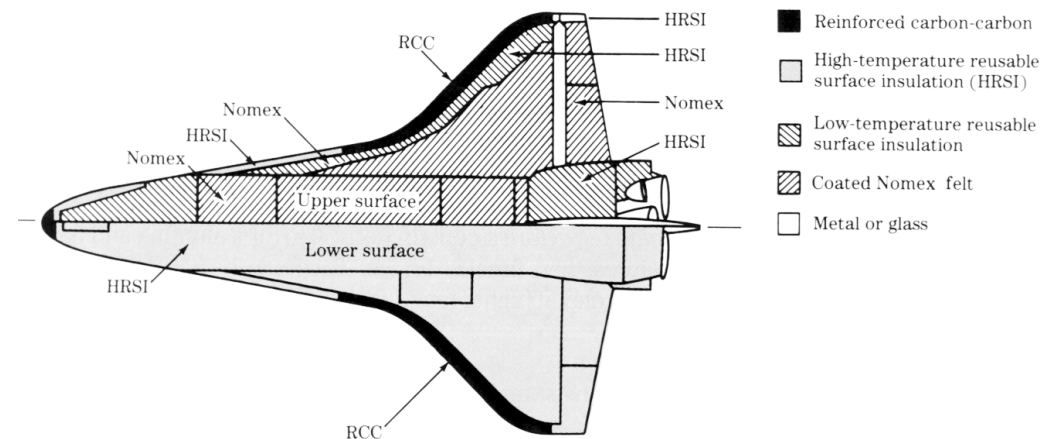
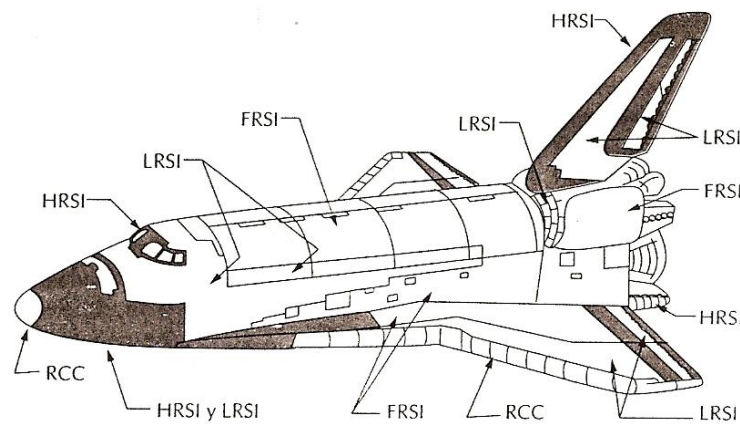


The **space shuttle** is covered with approximately **24,000 ceramic tiles**

APPLICATIONS OF ENGINEERING CERAMICS

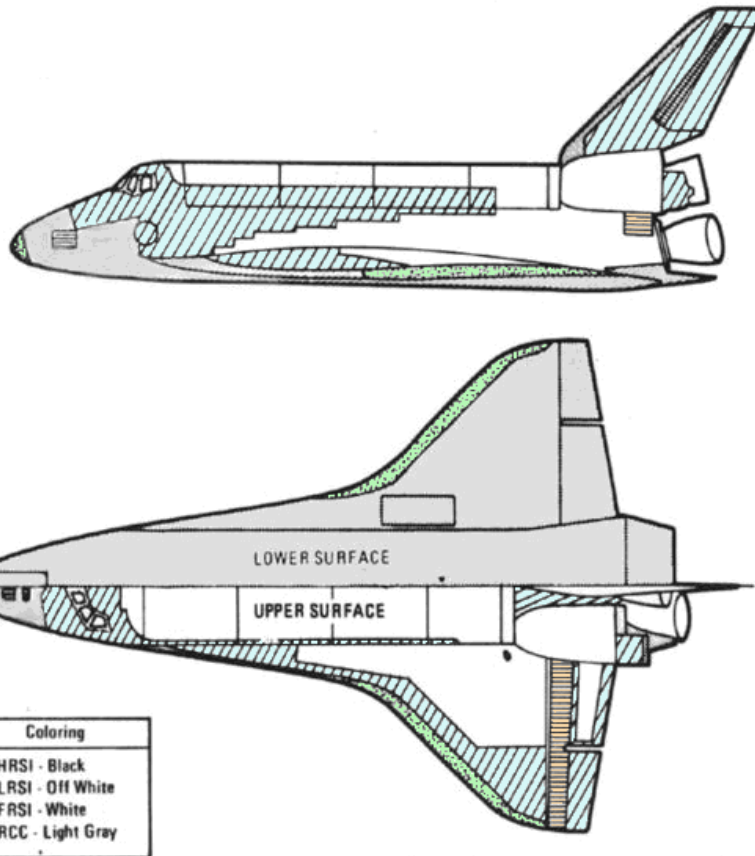
Thermal protection System for space shuttle

Name	Temp.	Material composition	Position	Color
Felt Reusable Surface Insulation (FRSI)	400°C	Aramid felt blankets (Nomex) coated with white-pigmented silicon elastomer coating	upper wing surfaces, upper payload bay doors, portions of the midfuselage and aft fuselage sides, portions of the upper wing surface	White flexible fabric (felt) (mostly replaced by FIB)
Low-temperature Reusable Surface Insulation (LRSI)	400-650	Tiles made from high purity silica fibers (white). Coated with borosilicate glass	upper fuselage, upper wing near the leading edge, selected areas of the forward, mid, and aft fuselage, vertical tail	white in color (by design in order to manage heat in direct sunlight)
High-temperature Reusable Surface Insulation (HRSI)	650-1260	Tiles made from high purity silica fibers (90% porous) (LI-900) (white). Coated with borosilicate glass with SiB_4 addition (black)	orbiter under surfaces, vertical stabilizer leading edge, areas on the upper forward fuselage	Black in color. Low thermal conductivity. (70% of surface area)
Reinforced Carbon Carbon (RCC)	1650	Pyrolytic carbon reinforced with carbon and coated with SiC	nose cap, wing leading edges	Light grey



APPLICATIONS OF ENGINEERING CERAMICS

Thermal protection System for space transporter



Thermal Protection System, Orbiter 102

	Reinforced Carbon-Carbon (RCC)
	High-Temperature, Reusable Surface Insulation (HRSI) —
	Low-Temperature, Reusable Surface Insulation (LRSI)
	Coated Nomex Felt Reusable Surface Insulation (FRSI)
	Metal or Glass

Element*	Area, sq m (sq ft)	Weight, kg (lb)
FRSI	332.7 (3581)	532.1 (1173)
LRSI	254.6 (2741)	1014.2 (2236)
HRSI	479.7 (5164)	4412.6 (9728)
RCC	38.0 (409)	1697.3 (3742)
Miscellaneous		918.5 (2025)
Total	1105.0 (11895)	8574.7 (18,904)

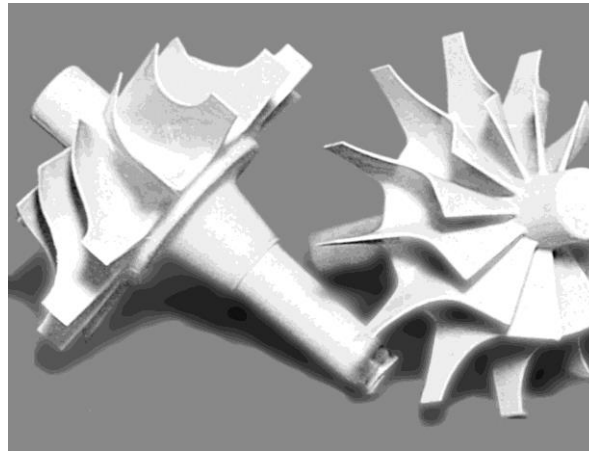
*Includes bulk insulation, thermal barriers, and closeouts



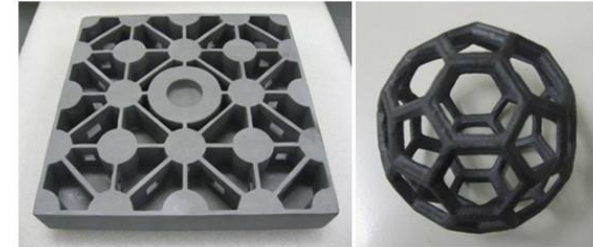
APPLICATIONS OF ENGINEERING CERAMICS

 Al_2O_3 , ZrO_2 , Si_3Ni_4 , SiC


High-strength alumina components for high-temperature applications

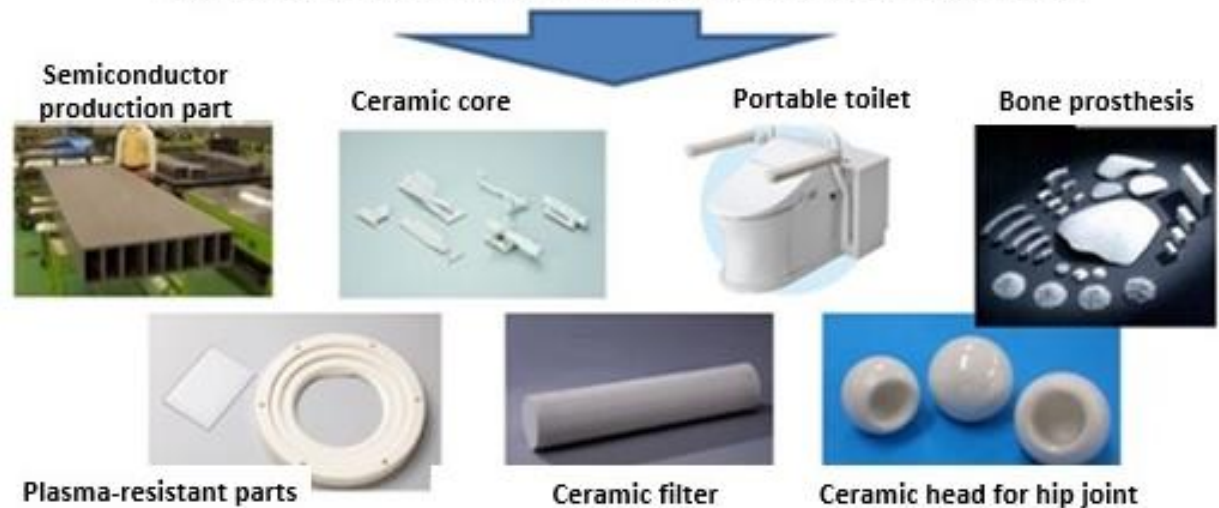


Gas-turbine rotors made of silicon nitride

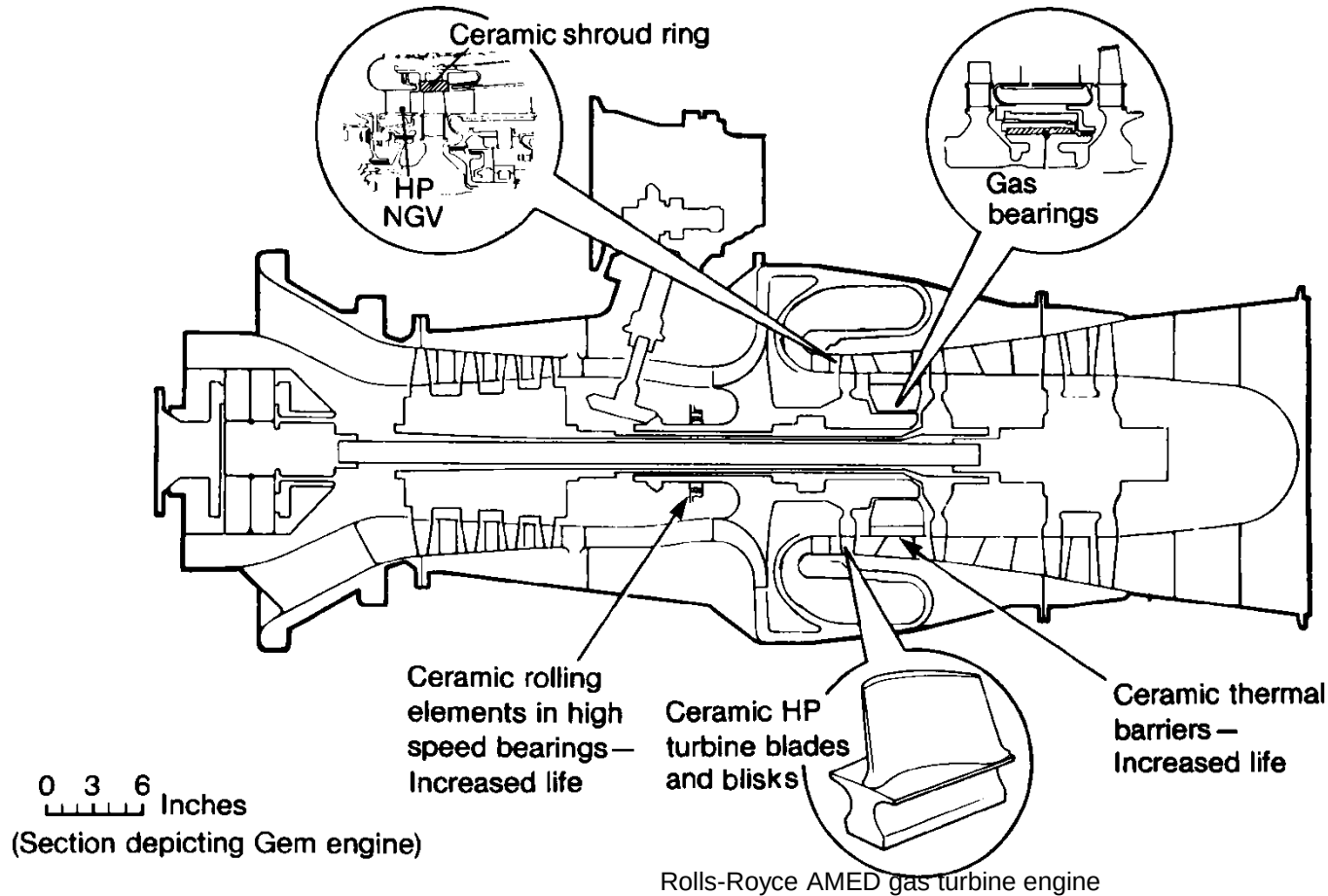


Ceramic members by 3D additive manufacturing

High-Value Added Ceramic Products Manufacturing Process



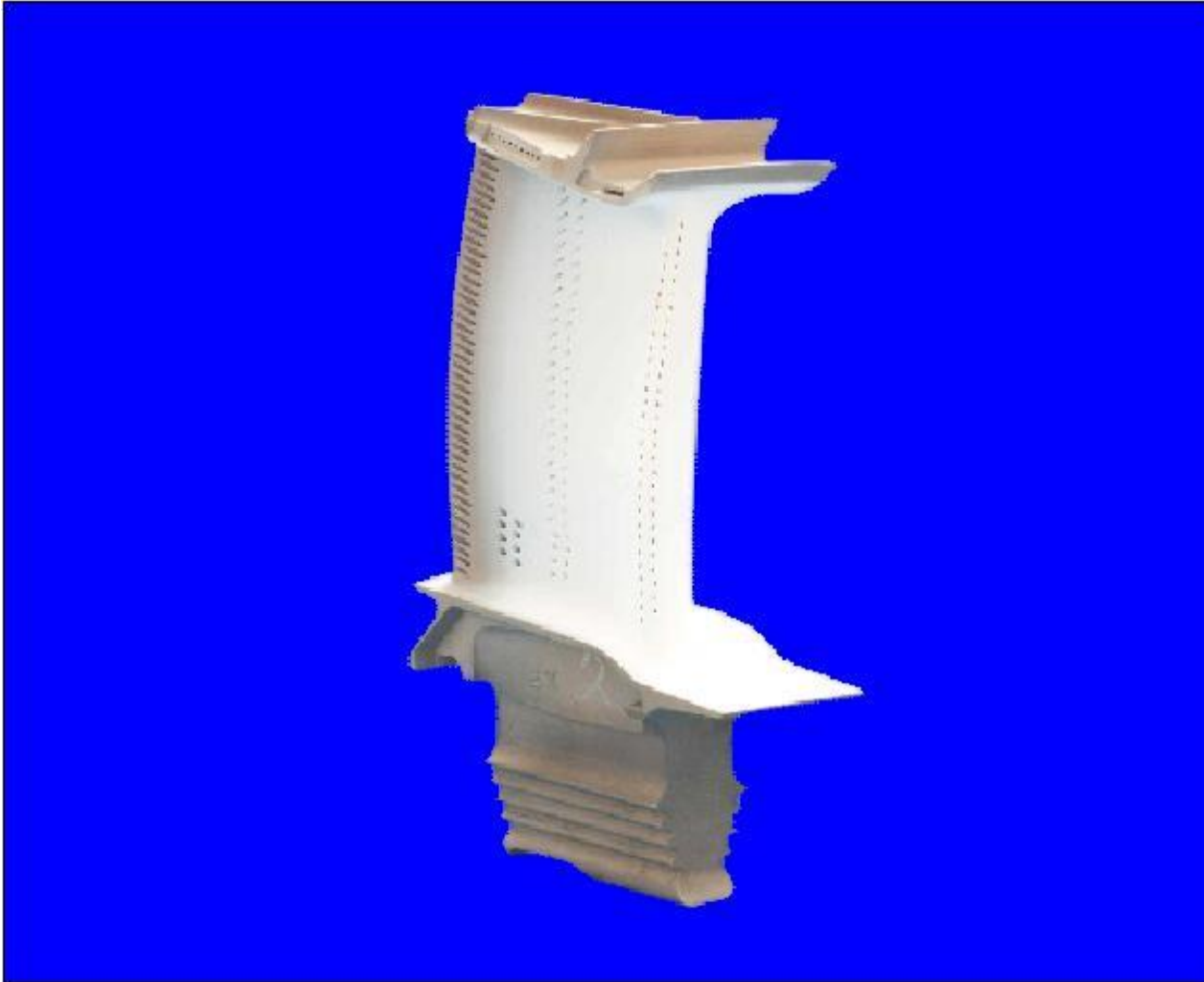
APPLICATIONS OF ENGINEERING CERAMICS



Combustor
 Support Struts
 Flameholder
 Re-heat shields
 Nozzle
 Turbine Airfoils
 Turbine Casing
 Exhaust Cone
 Jet Pipe

H.M. Flower, High Performance Materials in Aerospace, Chapman & Hall, 1995

APPLICATIONS OF ENGINEERING CERAMICS



Turbine Blade with Thermal Barrier coating (TBC)

ZrO_2 stabilized with Y_2O_3 (YSZ)

APPLICATIONS : TRADITIONAL CERAMICS

Clay: $\text{Al}_2\text{O}_3 \cdot \text{SiO}_2 \cdot \text{H}_2\text{O}$

Silica: SiO_2

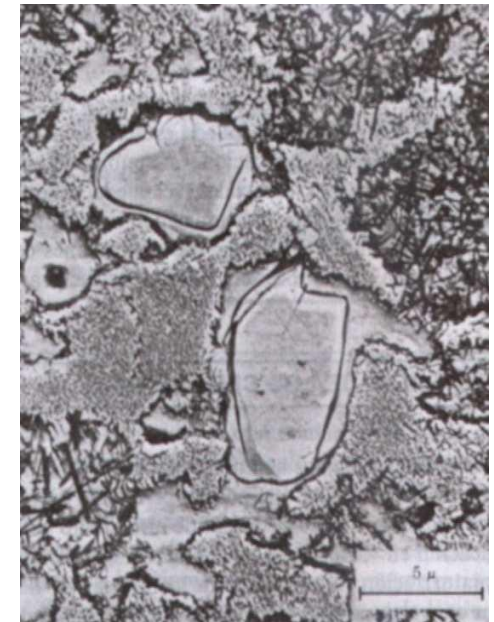
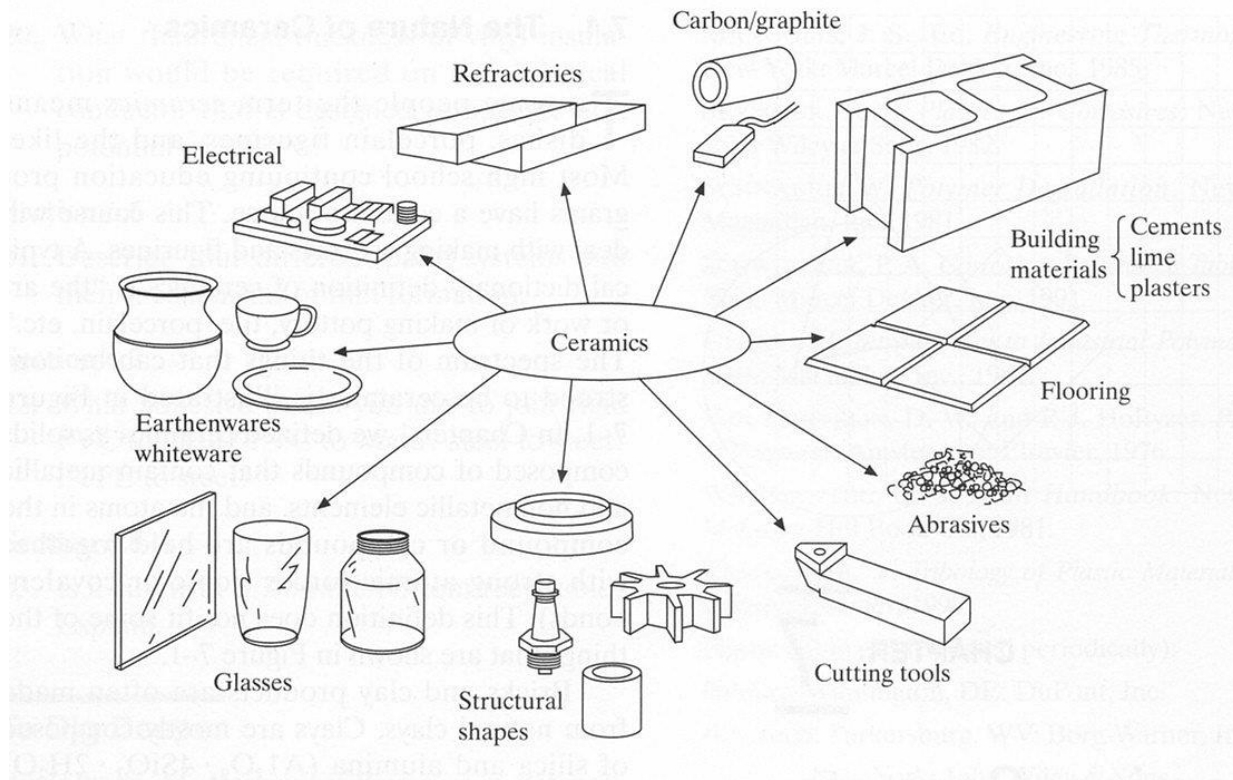
Feldspar: $\text{K}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2$

Electrical porcelain:

≈ 50% clay ($\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$)

≈ 25% SiO_2

≈ 25% feldspar ($\text{K}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 6\text{SiO}_2$).



Questions

1. How does the presence of defects (like porosity or cracks) introduced during manufacturing affect the fracture toughness of ceramics? Which manufacturing process would you choose to minimize such defects?
2. What manufacturing process would you select to produce a thin, flexible ceramic substrate with uniform thickness and smooth surfaces for use in electronic applications? Justify your choice.
3. If tasked with producing a complex-shaped ceramic part with high dimensional accuracy for biomedical implants (e.g., dental crowns), which process would you use to ensure both biocompatibility and precision? Why?
4. In which ceramic manufacturing processes is feedstock used, and what is the purpose of a binder in a feedstock?
5. What are the main differences between dry pressing and slip casting in ceramic processing
6. What are the key factors to consider when processing glass?
7. Can you explain the sintering process and its role in the consolidation and densification of ceramic materials?